Semiconductor

A semiconductor is a material that is made of silicon or germanium or gallium arsenide, whose electrical conductivity falls between conductor and insulator . Conductor like **copper** and Insulator like **glass**. It controls conductivity by allowing and restricting current flow by using external factors such as voltage, temperature. Semiconductors are very essential for us. It is playing an important role in the rapid growth of technology. For instance smartphones, laptops , tractors and satellites and missiles etc. We can cen change their conductivity by varying the temperature and adding impurities through doping, by creating **p - type** semiconductor with positive holes and **n- type** semiconductor with extra electrons, by which we made electronics devices such as chips , diodes, and transistors. which is most helpful in the growth of technology.

History

Semiconductor electronics is very much responsible for the growth of the whole world's technology. So now let's know what role has played the discovery of silicon or semimetals. How technology progressed from time to time. Semiconductor technology has become necessary for technology. It has also become the most necessary part of our lives and technical everyday lives. For instance, some devices such as laptops, smartphones, tablets and cars ect., we often use in our daily lives. Such as phones helping us in easily communicating with others by calling, sending messages through WhatsApp, Instagram and Facebook. Phones are often used in online payment instead of paying cash whenever we buy something or at another time. Their technology also helps in studying, Banking, marketing, etc. and cars used in travelling. So now let's know history about semiconductor electronics from the beginningto the present day and future.

The Discovery of Silicon

In the 19th century, The Swedish chemist Jones Jakob Berzelius who finally defined silicon in its pure form in 1824. and he was the person who gave its name i.e semimetal. That time it's become one of the defining semiconductor materials in electronics circuits. and also observed that its properties are as semiconductor and semimetal. That time it was not so important. And also they did not completely understand their importance. Early electrical engineers mostly used germanium as material instead of silicon.In 1874, Karl Ferdinand Braun is the person who discover rectifier effect. After that they understood the importance of semiconductor.And it became a very important and backbone of modern electronics devices. They used this to convert alternating current in Direct current. At the middle of the 20th century, semiconductor technology did not completely develop.

Transistor

In the beginning of 20th century, the three American scientists John Bardeen, Walter House Brattain and William Shockey developed a semiconductor that is Germanium's best device that leads the flow of electric current, they called it transistor. For this discovery, scientists were awarded the Nobel prize . Transistors were small in size and they consumed less current. It was widely used in electronic devices such as transistor radio. They have one disadvantage: they do not drive without power. After development of the field effect transistor, they become more powerful.

Microchip

In 1958, this was the first time when engineers allowed interconnecting multiple transistors on a single silicon, they called it a Wafer. They chips were small in size and it was cheaper to produce and it consumes less power. So they could easily produce in large quality. So first military and aerospace adopt them . After some years, finally they made microprocessor and microcontroller. In 1971 the first microprocessor finally come in the market that was Intel 4004. Especially this was used in computer technology. At the same time, the first microcontroller developed. This is used in organizing and optimizing computer systems. It combines CPU and other components on a single cheap and used as temperature and level controller. Alarm system and washing machine also work with this system.

Moore's Law

In 1965 Jordan Moora stated that the number of transistors on a microchip doubles in every two years and its cost becomes half at the same time. A transistor controls the flow of electricity on a chip which is used in electronic devices such as phones or computers. A computer or phone became more powerful and smaller after double the number of transistors.

The Future Technology

The next 20 years chief engineer will be moving from Mora more to more than Mora means the industrial will be moving from restless two days drinking to 3D integration where new materials and specialized architecture will be available in the future most of the work will be held by robots

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